

23/07/26

SIMATS Online Programming Lab

PYTHON PROGRAMMING

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Problem Solving2

Q1 Harshad number

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, 18 % 9 == 0. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

n=int(input("Enter a number:"))
temp=n
sum=0
while temp>0:
 digit=temp%10
 sum=sum+digit

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The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.
a = int(input("Enter first: ")) b = int(input("Enter second: "))
while b = 0: temp = a a = b b = temp % a
print("GCD is", a)

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

a=int(input("Enter first: "))
b=int(input("Enter second: "))
while b!=0:
 temp=b
 b=a%b
 a=temp
print("GCD is",a)

Test Input

6
3

Output

Enter first: Enter second: GCD is 3

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Complete the missing lines in the Python program below. The program finds all pairs of elements in a list that sum to a target value. Fill in all seven blanks.
def find_pairs(lst, target):
 seen = ____ pairs = []
 for num in ____:
 complement = target - ____
 if complement in ____:
 pairs.append((complement, num))
 ____add(num)
 return ____
data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = ____ (data, t)
print("Pairs:", result)

Your Answer

def find_pairs(lst, target):
 seen=set()
 pairs=[]
 for num in lst:
 complement=target-num

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Q4 Bitwise Operators

10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)

Your Answer

a=0b1101
b=0o25
c=0x1A
result = (a&b)|c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(bin(result)>> 2, result<< 1, ~result)

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else: high = mid - 1
return -1
data = [4,5,6,7,0,1,2]
t = input("Target: ")
result = rotated_binary_search(data, t)
print("Found at:", result)

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

1 def rotated_binary_search(arr, target):
2 low= 0
3 high= len(arr)-1
4 while low <= high:
5 mid = (low+high)//2
6 if arr[mid]== target:
7 return mid
8 if arr[low] <= arr[mid]:
9 if target >= arr[low] and target < arr[mid]:
10 high = mid-1
11 else:
12 low =mid+1

Test Input

0

Output

Target:
Found at: 4